

Logical Bilateralism - Proofs, Models and Applications

Session 3 – Non-classical frameworks

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Recap from yesterday: Two strands of bilateralism

- Warrant-based bilateralism: relevant conditions under which propositions can be warrantably asserted or denied
- Motivating a proof system of natural deduction with signed formulas for assertion and denial
- Coherence-based bilateralism: relevant conditions under which *collections* of propositions can be coherently asserted and/or denied together
- Motivating a specific kind of reading of sequent calculus
- **Both motivate a PTS approach for classical logic**

Outline and aims of today's session

Topic: Non-classical frameworks for bilateralism

- Taking proof and refutation as primitive
- Bilateral proof-theoretic presentations of some non-classical logics: FDE, N4, 2Int
- Bilateral Models

Revisiting the coordination principles

[A] bilateral formalization [of a logic] is distinctive in that it will also contain co-ordination principles[...].

Rumfitt 2008, p. 1059

[W]e want to draw attention to the fact that Rumfitt's logical system is seriously paralysed if the co-ordination principles do not hold for all sentences of the language [...].

Ferreira 2008, p. 1054

However:

- Neither is bilateralism necessarily tied to CL
 - Nor is bilateralism necessarily tied to *any* coordination principles
- ⇒ Without the coordination principles Rumfitt's system is equivalent to Nelson's constructive and paraconsistent logic N4 (Gibbard 2002)

Constructive bilateralism: Proof systems with two derivability relations

A constructive bilateralism

Not speech acts as the primary notions to act upon in PTS

- ⇒ Rather: proof - refutation, verification - falsification
- ⇒ Duality between different inferential relationships:
 - Usual consequence relation captures notion of *verification* from premises to conclusion
 - Dual counterpart captures notion of *falsification* from premises to conclusion

Taking proofs as primitive: The BHK interpretation

Brouwer-Heyting-Kolmogorov/Kreisel (BHK) interpretation

Gives an interpretation of complex formulas in terms of proof via the clauses i)-iv) below.

For atomic formulas it is assumed that we have an intuitive understanding of what constitutes a proof, for example, pencil and paper calculation serves as a proof of “ $27 \times 37 = 999$ ”.

- i) A proof of $A \wedge B$ is a pair $\langle d, e \rangle$, where d is a proof of A and e is a proof of B .
- ii) A proof of $A \vee B$ is a pair $\langle i, d \rangle$, where $i = 0$ and d is a proof of A , or $i = 1$ and d is a proof of B .
- iii) A proof of $A \rightarrow B$ is a function that maps any proof of A to a proof of B .
- iv) There is no proof of $\perp$.

Intuitionistic ND is sound under the BHK interpretation, insofar that any formula provable in it has a proof in the sense of the BHK interpretation.

Criticism of BHK-interpretation:

By, e.g., Johansson (1930), Nelson (1949), López-Escobar (1972):

- $\perp \rightarrow A$ not a constructive principle
- ⇒ A proof of $\perp \rightarrow A$ would be function mapping any proof of $\perp$ to a proof of A
- Negation in intuitionistic logic ($A \rightarrow \perp$) does not feel very constructive either

For example, if one accepts that there is no construction that proves an absurdity (as do most people) then a salient property of the construction π that proves “not- A ” is that when π is applied to a particular non-existent construction (namely a proof of A) it yields another non-existent construction!

López-Escobar (1972, p. 362)

Criticism of BHK-interpretation:

- IL focuses on the constructibility of truth, does not provide a correspondingly constructive notion of falsity
- Notion of a constructive proof of falsity vacuous
- Non-constructive because it does not require an explicit construction demonstrating the falsity of A
- With the right understanding of negation, double negation elimination and DeMorgan laws are unproblematic from a constructive perspective
- Next to disjunction property we should have constructible falsity (dual conjunction) property:
 - If $\vdash A \vee B$, then $\vdash A$ or $\vdash B$
 - If $\vdash \sim(A \wedge B)$, then $\vdash \sim A$ or $\vdash \sim B$

Example: Nelson's $\mathbf{N4}$ (Almukdad & Nelson 1984, Prawitz 1965)

Extend system of rules for minimal logic by the following:

$$\frac{A}{\sim\sim A} \sim\sim I \quad \frac{\sim\sim A}{A} \sim\sim E$$

$$\frac{\sim A}{\sim(A \wedge B)} \sim\wedge I_1 \quad \frac{\sim B}{\sim(A \wedge B)} \sim\wedge I_2 \quad \frac{\sim(A \wedge B) \quad \begin{array}{c} [\sim A] \\ \vdots \\ \overline{C} \end{array} \quad \begin{array}{c} [\sim B] \\ \vdots \\ \overline{C} \end{array}}{C} \sim\wedge E$$

$$\frac{\sim A \quad \sim B}{\sim(A \vee B)} \sim\vee I \quad \frac{\sim(A \vee B)}{\sim A} \sim\vee E_1 \quad \frac{\sim(A \vee B)}{\sim B} \sim\vee E_2$$

$$\frac{A \quad \sim B}{\sim(A \rightarrow B)} \sim\rightarrow I \quad \frac{\sim(A \rightarrow B)}{A} \sim\rightarrow E_1 \quad \frac{\sim(A \rightarrow B)}{\sim B} \sim\rightarrow E_2$$

Taking refutations as primitive: Extending the BHK interpretation

Implementation: Taking refutations as primitive

López-Escobar proposes another understanding of “not- A ”:

“In addition of having the concept of ‘a construction c proves a formula B ’ there is at hand the concept of ‘a construction d refutes a formula C ’ and there is a construction which refutes A .”

- ⇒ Having a primitive notion of refutation next to one for proof, in order to have a really constructive understanding of negation
- ⇒ Gives clauses analogous to the ones of the BHK interpretation to explain the meaning of “construction c refutes formula A ”
- ⇒ Gives a system that is sound under these clauses and equivalent to N4

Implementation: Taking refutations as primitive

López-Escobar's dual version of BHK clauses: logical connectives are interpreted in terms of a primitive and constructive notion of *refutation*.

FDE/N4:

- i)^d A refutation of $A \wedge B$ is a pair $\langle i, d \rangle$, where $i = 0$ and d is a refutation of A , or $i = 1$ and d is a refutation of B .
- ii)^d A refutation of $A \vee B$ is a pair $\langle d, e \rangle$, where d is a refutation of A and e is a refutation of B .
- iii)^d A refutation of $A \rightarrow B$ is a pair $\langle d, e \rangle$, where d is a proof of A and e is a refutation of B .
- iv)^d A refutation of $\sim A$ is a proof of A .

Plus, substituting intuitionistic clause for $\perp$ by:

- iv)_{con} A proof of $\sim A$ is a refutation of A .

Bilateral proof theory for non-classical logics: FDE, N4, 2Int

Bilateral FDE/N4

$$\begin{array}{ccc}
 (\Gamma'; \Delta') & (\Gamma''; \Delta'') & (\Gamma; \Delta) \\
 \vdots & \vdots & \vdots \\
 \frac{\overline{A}}{A \wedge B} & \frac{\overline{B}}{A \wedge B} & \frac{\overline{A \wedge B}}{A} \wedge E_{p_1} \quad \frac{\overline{A \wedge B}}{B} \wedge E_{p_2}
 \end{array}$$

$$\begin{array}{ccc}
 (\Gamma; \Delta) & (\Gamma; \Delta) & (\Gamma'; \Delta') \quad (\Gamma''; \Delta'', \llbracket A \rrbracket) \quad (\Gamma'''; \Delta''', \llbracket B \rrbracket) \\
 \vdots & \vdots & \vdots \quad \vdots \quad \vdots \\
 \frac{\overline{\overline{A}}}{A \wedge B} \wedge I_{r_1} & \frac{\overline{\overline{B}}}{A \wedge B} \wedge I_{r_2} & \frac{\overline{\overline{A \wedge B}} \quad \overline{\overline{C}} \quad \overline{\overline{C}}}{C} \wedge E_r
 \end{array}$$

- Single lines for proofs; double lines for refutations
- $(\Gamma; \Delta)$ for assumptions (Γ) and counterassumptions (Δ)
- $\llbracket \dots \rrbracket$ for possible discharge of assumptions and $\overline{\overline{\dots}}$ for possible discharge of counterassumptions

Bilateral FDE/N4

$$\begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \overline{A} \\
 \hline
 A \vee B \quad \vee I_{p_1}
 \end{array}
 \qquad
 \begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \overline{B} \\
 \hline
 A \vee B \quad \vee I_{p_2}
 \end{array}
 \qquad
 \begin{array}{c}
 (\Gamma'; \Delta') \quad ([A], \Gamma''; \Delta'') \quad ([B], \Gamma'''; \Delta''') \\
 \vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \\
 \overline{A \vee B} \qquad \qquad \overline{C} \qquad \qquad \overline{C} \\
 \hline
 C \quad \vee E_p
 \end{array}$$

$$\begin{array}{c}
 (\Gamma'; \Delta') \quad (\Gamma''; \Delta'') \\
 \vdots \qquad \qquad \vdots \\
 \overline{\overline{A}} \qquad \overline{\overline{B}} \\
 \hline
 \overline{\overline{A \vee B}} \quad \vee I_r
 \end{array}
 \qquad
 \begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \overline{\overline{A \vee B}} \\
 \hline
 A \quad \vee E_{r_1}
 \end{array}
 \qquad
 \begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \overline{\overline{A \vee B}} \\
 \hline
 B \quad \vee E_{r_2}
 \end{array}$$

Bilateral FDE/N4

$$\begin{array}{cccc}
 (\Gamma; \Delta) & (\Gamma; \Delta) & (\Gamma; \Delta) & (\Gamma; \Delta) \\
 \vdots & \vdots & \vdots & \vdots \\
 \frac{\overline{\overline{A}}}{\sim A} \sim I_p & \frac{\overline{\sim A}}{A} \sim E_p & \frac{\overline{\overline{A}}}{\sim A} \sim I_r & \frac{\overline{\sim A}}{A} \sim E_r \\
 \\
 ([A], \Gamma; \Delta) & (\Gamma'; \Delta') & (\Gamma''; \Delta'') & \\
 \vdots & \vdots & \vdots & \\
 \frac{\overline{B}}{A \rightarrow B} \rightarrow I_p & \frac{\overline{A \rightarrow B}}{B} \rightarrow E_p & \frac{\overline{A}}{A} \rightarrow E_p & \\
 \\
 (\Gamma'; \Delta') & (\Gamma''; \Delta'') & (\Gamma; \Delta) & (\Gamma; \Delta) \\
 \vdots & \vdots & \vdots & \vdots \\
 \frac{\overline{\overline{A}} \quad \overline{\overline{B}}}{A \rightarrow B} \rightarrow I_r & \frac{\overline{\overline{A \rightarrow B}}}{A} \rightarrow E_{r_1} & \frac{\overline{\overline{A \rightarrow B}}}{B} \rightarrow E_{r_2} &
 \end{array}$$

Another example: Bi-intuitionistic logic 2Int (Wansing 2016, 2017)

- Bi-intuitionistic logic with same connectives as Heyting-Brouwer logic (Rauszer 1974): $\top, \perp, \wedge, \vee, \rightarrow^+, \rightarrow^-$
 - Co-implication: connective dual to implication but understanding in 2Int differs from the understanding in H-B-logic
 - BiInt: co-implication captures preservation of non-truth from conclusion to premises
 - 2Int: co-implication captures preservation of falsity from premises to conclusion
- ⇒ Against unilateralist objection that when considering refutation rules we have to rely on concept of proof (i.e., this is the more primitive concept)

Implication and co-implication in 2Int

- i)^d A refutation of $A \wedge B$ is a pair $\langle i, d \rangle$, where $i = 0$ and d is a refutation of A , or $i = 1$ and d is a refutation of B .
- ii)^d A refutation of $A \vee B$ is a pair $\langle d, e \rangle$, where d is a refutation of A and e is a refutation of B .
- iii)^d A refutation of $A \rightarrow^+ B$ is a pair $\langle d, e \rangle$, where d is a proof of A and e is a refutation of B .

$$\begin{array}{c}
 ([A], \Gamma; \Delta) \\
 \vdots \\
 \overline{B} \\
 \hline
 A \rightarrow^+ B \rightarrow^+ I_p
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \quad (\Gamma'; \Delta') \\
 \vdots \quad \vdots \\
 \overline{A} \quad \overline{A \rightarrow^+ B} \\
 \hline
 B \rightarrow^+ E_p
 \end{array}$$

$$\begin{array}{c}
 (\Gamma; \Delta) (\Gamma'; \Delta') \\
 \vdots \quad \vdots \\
 \overline{A} \quad \overline{B} \\
 \hline
 A \rightarrow^+ B \rightarrow^+ I_r
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \overline{\overline{A \rightarrow^+ B}} \\
 \hline
 A \rightarrow^+ E_{r1}
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \overline{\overline{A \rightarrow^+ B}} \\
 \hline
 B \rightarrow^+ E_{r2}
 \end{array}$$

Implication and co-implication in 2Int

$$\begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \hline
 \overline{B} \\
 \hline
 A \rightarrow^+ B \rightarrow^+ I_p
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \quad (\Gamma'; \Delta') \\
 \vdots \quad \vdots \\
 \hline
 \overline{A} \quad A \rightarrow^+ B \\
 \hline
 B \rightarrow^+ E_p
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta)(\Gamma'; \Delta') \\
 \vdots \quad \vdots \\
 \hline
 \overline{A} \quad \overline{B} \\
 \hline
 A \rightarrow^+ B \rightarrow^+ I_r
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \hline
 \overline{A \rightarrow^+ B} \\
 \hline
 A \rightarrow^+ E_{r1}
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \hline
 \overline{A \rightarrow^+ B} \\
 \hline
 B \rightarrow^+ E_{r2}
 \end{array}$$

$$\begin{array}{c}
 (\Gamma; \Delta, \llbracket A \rrbracket) \\
 \vdots \\
 \hline
 \overline{\overline{B}} \\
 \hline
 A \rightarrow^- B \rightarrow^- I^d
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \quad (\Gamma'; \Delta') \\
 \vdots \quad \vdots \\
 \hline
 \overline{A \rightarrow^- B} \quad \overline{A} \\
 \hline
 B \rightarrow^- E^d
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma'; \Delta') \quad (\Gamma; \Delta) \\
 \vdots \quad \vdots \\
 \hline
 \overline{A} \quad \overline{B} \\
 \hline
 A \rightarrow^- B \rightarrow^- I
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \hline
 \overline{A \rightarrow^- B} \\
 \hline
 B \rightarrow^- E_1
 \end{array}
 \quad
 \begin{array}{c}
 (\Gamma; \Delta) \\
 \vdots \\
 \hline
 \overline{A \rightarrow^- B} \\
 \hline
 A \rightarrow^- E_2
 \end{array}$$

⇒ When co-implication is part of the system the proof rules are based just the same on the concept of refutation

- Both can be understood to express dual entailment in the object language:
 - proof of $A \rightarrow^+ B$ iff there is a proof from A to B ; $\vdash^+ A \rightarrow^+ B$ iff $(A; \emptyset) \vdash^+ B$
 - refutation of $A \rightarrow^- B$ iff there is a refutation from A to B ; $\vdash^- A \rightarrow^- B$ iff $(\emptyset; A) \vdash^- B$

From bilateral natural deduction to sequent calculus

Equivalent SC system to N2Int: SC2Int (Ayhan 2020)

- Bilateral cut-free (G3-style) sequent calculus with two consequence relations
- Sequents of the form $(\Gamma; \Delta) \vdash^* C$ ($*$ = $+$ or $-$)

Rules for $\rightarrow$ in SC2Int:

$$\frac{(\Gamma, A; \Delta) \vdash^+ B}{(\Gamma; \Delta) \vdash^+ A \rightarrow B} \rightarrow R^+ \qquad \frac{(\Gamma, A \rightarrow B; \Delta) \vdash^+ A \quad (\Gamma, B; \Delta) \vdash^* C}{(\Gamma, A \rightarrow B; \Delta) \vdash^* C} \rightarrow L^a$$

$$\frac{(\Gamma; \Delta) \vdash^+ A \quad (\Gamma; \Delta) \vdash^- B}{(\Gamma; \Delta) \vdash^- A \rightarrow B} \rightarrow R^- \qquad \frac{(\Gamma, A; \Delta, B) \vdash^* C}{(\Gamma; \Delta, A \rightarrow B) \vdash^* C} \rightarrow L^c$$

$\rightarrow$ Single lines from ND translate into $\vdash^+$ and double lines into $\vdash^-$

Proof and refutation rules in Sequent Calculus

For $* \in \{+, -\}$:

Zero-premise rules:

$$\frac{}{(\Gamma, p; \Delta) \vdash^+ p} Rf^+$$

$$\frac{}{(\Gamma; \Delta, p) \vdash^- p} Rf^-$$

$$\frac{}{(\Gamma; \Delta) \vdash^- \perp} \perp R^-$$

$$\frac{}{(\Gamma, \perp; \Delta) \vdash^* C} \perp L^a$$

$$\frac{}{(\Gamma; \Delta) \vdash^+ \top} \top R^+$$

$$\frac{}{(\Gamma; \Delta, \top) \vdash^* C} \top L^c$$

Proof and refutation rules in Sequent Calculus

The following structural rules can shown to be admissible in SC2Int:

$$\frac{(\Gamma; \Delta) \vdash^* C}{(\Gamma, A; \Delta) \vdash^* C} W^a \quad \frac{(\Gamma; \Delta) \vdash^* C}{(\Gamma; \Delta, A) \vdash^* C} W^c$$

$$\frac{(\Gamma, A, A; \Delta) \vdash^* C}{(\Gamma, A; \Delta) \vdash^* C} C^a \quad \frac{(\Gamma; \Delta, A, A) \vdash^* C}{(\Gamma; \Delta, A) \vdash^* C} C^c$$

$$\frac{(\Gamma; \Delta) \vdash^+ D \quad (\Gamma', D; \Delta') \vdash^* C}{(\Gamma, \Gamma'; \Delta, \Delta') \vdash^* C} Cut^a \quad \frac{(\Gamma; \Delta) \vdash^- D \quad (\Gamma'; \Delta', D) \vdash^* C}{(\Gamma, \Gamma'; \Delta, \Delta') \vdash^* C} Cut^c$$

Non-classical Bilateral Models

First Degree Entailment: FDE

Take a relation ρ on $\text{Atoms} \times \{0, 1\}$.

Extend it to the whole language like this:

- $\perp \rho 1, \perp \rho 0$
- $\sim A \rho 1$ iff $A \rho 0$
- $\sim A \rho 0$ iff $A \rho 1$
- $A \wedge B \rho 1$ iff $A \rho 1$ and $B \rho 1$
- $A \wedge B \rho 0$ iff $A \rho 0$ or $B \rho 0$
- $A \vee B \rho 1$ iff $A \rho 1$ or $B \rho 1$
- $A \vee B \rho 0$ iff $A \rho 0$ and $B \rho 0$.

This is *bilateral*: truth and falsity are equally fundamental.

They are *independent*: ρ allows for *gaps* and *gluts*.

Validity Notions

- $(\Gamma; \Delta) \models^+ C$ iff for each ρ , if $A \rho 1$ for each $A \in \Gamma$ and $B \rho 0$ for each $B \in \Delta$, then $C \rho 1$.
- $(\Gamma; \Delta) \models^- C$ iff for each ρ , if $A \rho 1$ for each $A \in \Gamma$ and $B \rho 0$ for each $B \in \Delta$, then $C \rho 0$.

Examples

Choose a ρ where $p \rho 1, p \rho 0, q \not\rho 1$ and $q \not\rho 0$.

This valuation shows that

$$(p, \sim p;) \not\models^+ q \text{ and } (p;) \not\models^+ q \vee \sim q$$

Heredity

Important fact: If $\rho \subseteq \rho'$ on *atoms*,

(i.e., for every atom p , and $i \in \{0, 1\}$, if $p \rho i$ then $p \rho' i$ too)

then $\rho \subseteq \rho'$ on *all formulas*.

(i.e., for every formula A , and $i \in \{0, 1\}$, if $A \rho i$ then $A \rho' i$ too)

Shown by an easy induction on the construction of A .

This relies on the specific behaviour of the connectives.

Soundness and Completeness

To prove soundness (for natural deduction or sequents), use an induction on the construction of the proof.

Completeness is more involved, and I will defer this to our discussion of models for $N4$ and $2Int$.

Restricting to just Gaps or Gluts

Relations that are partial functions on atoms (if $p \rho 0$ then $p \not\rho 1$)
are partial functions on the whole language, too.

If we restrict ourselves to these, we have K3, Kleene's strong three-valued logic. $(p \wedge \sim p;) \models_{K3}^+ q$

Relations that are total on atoms (either $p \rho 0$ or $p \rho 1$)
are total on the whole language, too.

If restrict ourselves to these, we have LP, Priest's *logic of paradox*.
 $(;) \models_{LP} p \vee \sim p$

Extending Models to N4 and 2Int

We relativise valuations to a set of points, partially ordered by a relation $\sqsubseteq$.

We impose the natural heredity condition: if $w \sqsubseteq w'$ then $\rho_w \subseteq \rho_{w'}$ (on atoms).

The evaluation clauses for $\perp, \wedge, \vee, \sim$ are as before.

- $A \rightarrow B \rho_w 0$ iff $A \rho_w 1$ and $B \rho_w 0$.
- $A \rightarrow B \rho_w 1$ iff $\forall w' \sqsupseteq w$, if $A \rho_{w'} 1$ then $B \rho_{w'} 1$.
- $A \rightarrow^- B \rho 1$ iff $A \rho_w 0$ and $B \rho_w 1$.
- $A \rightarrow^- B \rho 0$ iff $\forall w' \sqsubseteq w$, if $A \rho_{w'} 0$ then $B \rho_{w'} 0$.

These clauses are thoroughly bilateral. Positive and negative information are equally constrained by the partial order on points.

Heredity Reloaded

If $w \sqsubseteq w'$ then $\rho_w \subseteq \rho_{w'}$ on atoms.

- $A \rightarrow B \rho_w 0$ iff $A \rho_w 1$ and $B \rho_w 0$.
- $A \rightarrow B \rho_w 1$ iff $\forall w' \sqsupseteq w$, if $A \rho_{w'} 1$ then $B \rho_{w'} 1$.
- $A \rightarrow^- B \rho_w 1$ iff $A \rho_w 0$ and $B \rho_w 1$.
- $A \rightarrow^- B \rho_w 0$ iff $\forall w' \sqsubseteq w$, if $A \rho_{w'} 0$ then $B \rho_{w'} 0$.

If $w \sqsubseteq w'$ then $\rho_w \subseteq \rho_{w'}$ over the whole language.

(This is why we impose the universal quantifier clause on the positive $\rightarrow$ and negative $\rightarrow^-$ conditions.)

Making Models

Any Kripke model for intuitionistic propositional logic can be used to generate a number of $\mathbf{N4}$ and $\mathbf{2Int}$ models.

Notice that the *truth* ($\rho_w \vdash$) conditions for $\perp$, $\wedge$, $\vee$ and $\rightarrow$ agree with the usual intuitionistic truth conditions.

(These truth do not involve checking the falsity status of their components.)

So, we can extend any Kripke frame for intuitionistic logic into an $\mathbf{N4}$ or $\mathbf{2Int}$ model by using the model's forcing relation for *truth*, and by imposing whatever conditions we like for *falsity*.

The bilateral model's notion of *truth* agrees with the original model's support relation on the fragment including $\perp$, $\wedge$, $\vee$ and $\rightarrow$.

Non-classical Bilateralism for Cheap

This construction has a syntactic analogue. We can embed $\mathsf{N4}$ and $\mathsf{2Int}$ inside intuitionistic logic as follows.

Extend the language of intuitionistic logic with a new family of atoms: a distinct $\bar{p}$ for each original atom, p .

We can use $\bar{}$ to model the *falsity* conditions for atoms, and thereby, the falsity conditions for *all* $\mathsf{N4}$ and $\mathsf{2Int}$ formulas, inside unilateral intuitionist models.

Non-Classical Bilateralism for Cheap

We translate $\mathbf{N4}$ or $\mathbf{2Int}$ into the language of intuitionistic logic, using two translation functions, t and f

- $t(\top) = \top, f(\top) = \perp; \quad t(\perp) = \perp, f(\perp) = \top.$
- $t(p) = p, f(p) = \bar{p}.$
- $t(A \wedge B) = t(A) \wedge t(B), f(A \wedge B) = f(A) \vee f(B).$
- $t(A \vee B) = t(A) \vee t(B), f(A \vee B) = f(A) \wedge f(B).$
- $t(A \rightarrow B) = t(A) \rightarrow t(B), f(A \rightarrow B) = t(A) \wedge f(B).$
- $t(\sim A) = f(A), f(\sim A) = t(A).$
- $t(A \rightarrow^- B) = f(A) \wedge t(B), f(A \rightarrow^- B) = f(A) \rightarrow f(B)$

Example:

$$t((p \rightarrow q) \rightarrow^- \sim p) = f(p \rightarrow q) \wedge t(\sim p) = (t(p) \wedge f(q)) \wedge f(p) = (p \wedge \bar{q}) \wedge \bar{p}$$

Non-Classical Bilateralism for Cheap

Using t and f we can prove the following correspondences:

$$(\Gamma; \Delta) \models_{\text{N4}/2\text{Int}}^+ C \text{ iff } t(\Gamma), f(\Delta) \models_{\text{Int}} t(C)$$

$$(\Gamma; \Delta) \models_{\text{N4}/2\text{Int}}^- C \text{ iff } t(\Gamma), f(\Delta) \models_{\text{Int}} f(C)$$

From any $\text{N4}/2\text{Int}$ counterexample, construct an intuitionistic counterexample to the translation by using the bilateral falsity conditions to define the required truth conditions for the $\bar{p}$ atoms.

Conversely, from any Int counterexample, use the truth conditions for the $\bar{p}$ atoms to define the falsity conditions for each atom.

The structure of the t and f translations ensure that the models line up appropriately.

Fun project: State and prove the appropriate proof-theoretic analogue of this result, in natural deduction or the sequent calculus (or better, both!).

Restricting to just Gaps

If, in a model, ρ_w is a partial function on atoms for each point w , then indeed, it remains a partial function on the entire language.

If $A \rightarrow B \ \rho_w \ 1$ then $\forall w' \sqsubseteq w$, if $A \ \rho_{w'} \ 1$ then $B \ \rho_{w'} \ 1$.

If $A \rightarrow B \ \rho_w \ 0$ then $A \ \rho_w \ 1$ and $B \ \rho_w \ 0$.

So if both hold, since $w \sqsubseteq w$, we must have $B \ \rho_w \ 0$ and $B \ \rho_w \ 1$.

If $A \rightarrow^- B \ \rho_w \ 0$ then $\forall w' \sqsubseteq w$, if $A \ \rho_{w'} \ 0$ then $B \ \rho_{w'} \ 1$.

If $A \rightarrow^- B \ \rho_w \ 1$ then $A \ \rho_w \ 0$ and $B \ \rho_w \ 1$.

And again, since $w \sqsubseteq w$, we must have $B \ \rho_w \ 1$ and $B \ \rho_w \ 0$.

So, if a glut enters the language, it must do so at the atoms.

Restricting to Gluts fails

The analogous result does *not* hold for total valuations (those with no gaps).

Take a two-point model on w, v where $w \sqsubset v$.

Set $p, q \rho_w 0$ and $p, q \not\rho_w 1$. Add $p \rho_v 1$.

(So p and q are false everywhere, and p is also true at v .)

So, ρ is total on the atoms p and q at both points.

However, $p \rightarrow q \not\rho_w 0$, since $p \not\rho_w 1$,

and $p \rightarrow q \not\rho_w 1$, since $v \sqsubseteq w, p \rho_v 1$ and $q \not\rho_v 1$.

Even though there are no gaps on the *atoms*, a gap has entered the language by way of the conditional.

Positions for Non-Classical Bilateralist Logics

I will just *sketch* a technique we can use to prove completeness, generalising the notion of a position we introduced on Day 1.

Suppose we cannot derive $(\Gamma; \Delta) \vdash^* C$.

We want a model with a point w at which every member of Γ is true, every member of Δ is false and where C fails to be true (if $* = +$) or fails to be false (if $* = -$).

The strategy of using *positions* is to make the model out of underivable sequents.

Bilateral Positions: Take 1

The pair of pairs $[(\mathcal{X}; \mathcal{X}') : (\mathcal{Y}; \mathcal{Y}')]$ of sets of formulas is said to be *available* iff for every finite Γ taken from $\mathcal{X}$ and Δ taken from $\mathcal{X}'$, there is no $C \in \mathcal{Y}$ where we can prove $(\Gamma; \Delta) \vdash^+ C$, and no $D \in \mathcal{Y}'$ where we can prove $(\Gamma; \Delta) \vdash^- D$.

Defn: $[(\mathcal{X}_2; \mathcal{X}'_2) : (\mathcal{Y}_2; \mathcal{Y}'_2)]$ *extends* $[(\mathcal{X}_1; \mathcal{X}'_1) : (\mathcal{Y}_1; \mathcal{Y}'_1)]$ iff $\mathcal{X}_1 \subseteq \mathcal{X}_2$, $\mathcal{X}'_1 \subseteq \mathcal{X}'_2$, $\mathcal{Y}_1 \subseteq \mathcal{Y}_2$ and $\mathcal{Y}'_1 \subseteq \mathcal{Y}'_2$.

Fact: Any available position can be extended to an available position $[(\mathcal{X}_*; \mathcal{X}'_*); (\mathcal{Y}_*; \mathcal{Y}'_*)]$ where $\mathcal{X}_*, \mathcal{Y}_*$ partition the language, and $\mathcal{X}'_*, \mathcal{Y}'_*$ also partition the language. **Why?** Both forms of *Cut* are what you need to show that each formula can be added into one side of the $\mathcal{X}, \mathcal{Y}$ partition, and also to one side of the $\mathcal{X}', \mathcal{Y}'$ partition.

The available partitions can be used to make a model of N4 or 2Int, but they are not *our* models. $[(p \vee q;) : (p, q;)]$ is available, which means we have points at which $p \vee q$ is true, but p and q both fail to be true.

Bilateral Positions: Take 2

To construct partitions that match our models for $\mathbf{N4}$ and $\mathbf{2Int}$, we use a stricter criterion. We say that $[(\mathcal{X}; \mathcal{X}') : (\mathcal{Y}; \mathcal{Y}')] is *Kripke* available iff for every finite Γ taken from $\mathcal{X}$ and Δ taken from $\mathcal{X}'$, there is no C that is a disjunction of formulas taken from $\mathcal{Y}$ and negations of formulas taken from $\mathcal{Y}'$ where we have $(\Gamma; \Delta) \vdash^+ C$.$

(Or equivalently there is no D that is a conjunction of formulas taken from $\mathcal{Y}'$ and negations of formulas taken from $\mathcal{Y}$ where we have $(\Gamma; \Delta) \vdash^- D$.)

Every Kripke available position may be extended to a partition, as before.

It is a routine exercise to show that the Kripke available partition positions, ordered under *strengthening* (where $[(\mathcal{X}_2; \mathcal{X}'_2) : (\mathcal{Y}_2; \mathcal{Y}'_2)]$ strengthens $[(\mathcal{X}_1; \mathcal{X}'_1) : (\mathcal{Y}_1; \mathcal{Y}'_1)]$ when $\mathcal{X}_1 \subseteq \mathcal{X}_2$ and $\mathcal{X}'_1 \subseteq \mathcal{X}'_2$) form a model for $\mathbf{N4}$ or $\mathbf{2Int}$. Here, the formulas in $\mathcal{X}$ are *true* and those in $\mathcal{X}'$ are *false* in $[(\mathcal{X}; \mathcal{X}') : (\mathcal{Y}; \mathcal{Y}')]$. This gives us a *completeness* proof.

Summary

- We have explored the non-classical bilateral logics FDE, N4 and 2Int from different perspectives.
- We have examined a number of different natural deduction systems:
 - one in which negation plays a central role in expressing the relation between truth and falsity conditions,
 - and another, in which proof and refutation are equally primitive structural features of proof.
- This second system corresponds to a sequent calculus with two kinds of sequent, for proofs and refutations, and a context with two different kinds of assumption.
- We ended with a development of a properly bilateral *relational* valuation extension of traditional Kripke semantics for intuitionistic logic.

Tomorrow

We explore ways to bilateralise the λ calculus.

Thanks for your attention!

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